

ROS2 ATOSFleetManagement Cheatsheet

Updated: 2026-04-23

1) Build

Option A (recommended helper script)

```
cd <ATOS_REPO>
./scripts/build_atosfleetmanagement.sh
```

Option B (explicit colcon)

```
cd <ATOS_WS>
source /opt/ros/humble/setup.bash
colcon build --packages-select atos atos_gui --symlink-install
source install/setup.bash
```

2) Run ATOSFleetManagement

Without simulators

```
cd <ATOS_WS>
source /opt/ros/humble/setup.bash
source install/setup.bash
ros2 launch atos launch_atosfleetmanagement.py insecure:=True with_truck_simulator:=False
```

With simulators

```
cd <ATOS_WS>
source /opt/ros/humble/setup.bash
source install/setup.bash
ros2 launch atos launch_atosfleetmanagement.py insecure:=True with_truck_simulator:=True
```

GUI:

- URL: `http://localhost:8420`
- Hard refresh after GUI JS updates: `Ctrl+Shift+R`

3) Runtime control for simulators

Stop SIM-2 while launch keeps running

```
pkill -f "__node:=atos_truck_simulator_2"
```

Restart SIM-2 manually (same parameters as launch)

```
cd <ATOS_WS>
source /opt/ros/humble/setup.bash
source install/setup.bash
ros2 run atos atos_truck_simulator --ros-args \
  -r __ns:=/atos \
  -r __node:=atos_truck_simulator_2 \
  -p uid:=L5S-TRUCK-SIM-2 \
  -p start_index:=250 \
  -p target_speed_kmh:=40.0 \
  -p acceleration_mps2:=2.0
```

4) Testing and diagnostics

Check truck state topic once

```
ros2 topic echo /atos/truck_objects/state --once
```

Look for fields:

- `last\_cot\_message`
- `last\_tcp\_command`
- `last\_tcp\_warning`
- `tcp\_connected`
- `path\_name`, `path\_index`, `speed\_mps`

Expected UI behavior

- Map size fixed to `600 x 650`.
- Right panel shows truck table.
- Each row contains:
 - latest CoT message block (scrollable)
 - last TCP command block (scrollable)
 - red TCP warning text when send/disconnect issues occur

Typical TCP warnings in UI

- `TCP client disconnected`
- `No active TCP socket for this truck`
- `TCP send failed (errno=...)`
- `TLS send failed (ssl\_error=...)`

5) Control command semantics (TruckObjectControl -> Truck)

Per-truck TCP command field command= values:

- `STOP`: set speed to stop target (`target\_speed\_mps=0` by default)
- `SLOWDOWN`: set speed to warning target (`30 km/h` default => `8.333... m/s`)
- `DRIVE`: release speed control back to truck/pathfollower behavior

Typical TCP payload example:

```
command=DRIVE;target_speed_mps=nochange;distance_to_truck_ahead_m=651.35;truck_ahead_path_index=254;truck_ahead_path_name=...
```

6) Notes on speed units

- Simulators send CoT speed in **km/h**.
- TruckObjectControl internally uses **speed_mps**.
- GUI displays speed in **km/h**.

7) Quick rebuild after code changes

```
cd <ATOS_WS>
source /opt/ros/humble/setup.bash
colcon build --packages-select atos atos_gui --symlink-install
source install/setup.bash
```

8) Minimal local example paths

- `` example: `/home/sepast/Documents/repos/ATOS`
- `` example: `/home/sepast/atos\_ws`